

12-Week 3-Hour Daily DSA Routine for a 50 LPA Offer

Introduction

This comprehensive 12-week Data Structures and Algorithms (DSA) routine is designed to prepare you for technical interviews at top-tier tech companies offering packages around 50 Lakhs Per Annum (LPA). The routine assumes basic programming knowledge in a language like Python, C++, or Java and is structured for 3 hours of daily study, split into learning, practice, and revision. By following this plan consistently, you will build a strong foundation in DSA, develop problem-solving skills, and gain confidence to tackle high-stakes interviews. The routine covers essential topics, provides daily tasks, and includes resources and strategies to maximize your preparation.

1 Routine Overview

The 3-hour daily schedule is divided as follows:

- **Learning (1 hour):** Study core DSA concepts, algorithms, and techniques through resources like books, articles, or videos.
- **Practice (1.5 hours):** Solve 2–3 problems daily to apply concepts and improve coding efficiency.
- **Revision (0.5 hours):** Review previously solved problems and concepts to reinforce learning and identify gaps.

The 12-week plan covers key DSA topics: arrays, strings, linked lists, stacks, queues, trees, graphs, dynamic programming, and advanced topics like tries and bit manipulation. Each week includes specific learning goals, practice problems, and revision tasks.

2 12-Week Detailed Plan

Below is a week-by-week breakdown of the DSA routine, with detailed daily tasks for each topic.

2.1 Week 1: Arrays

2.1.1 Learning Objectives

Understand array manipulation, two-pointer technique, and sliding window. Learn time and space complexity analysis.

System:]□□

System:

- Book: “Introduction to Algorithms” by Cormen (Chapter 10)
- Online: GeeksforGeeks (Arrays section), NeetCode YouTube
- Platform: LeetCode (Array problems)

2.1.2 Daily Breakdown

- **Day 1:** Learn array basics and two-pointer technique. Solve “Two Sum” (LeetCode Easy).
- **Day 2:** Study sliding window. Solve “Maximum Subarray” (LeetCode Easy).
- **Day 3:** Learn prefix sum technique. Solve “Subarray Sum Equals K” (LeetCode Medium).
- **Day 4:** Practice array rotation. Solve “Rotate Array” (LeetCode Medium).
- **Day 5:** Study merging intervals. Solve “Merge Intervals” (LeetCode Medium).
- **Day 6:** Mixed practice. Solve 2–3 array problems (Easy/Medium).
- **Day 7:** Revise all Week 1 problems. Analyze time/space complexity for each.

2.2 Week 2: Strings

2.2.1 Learning Objectives

Master string manipulation, pattern matching (KMP, Rabin-Karp), and string-specific algorithms.

2.2.2 Resources

- Book: “Algorithms” by Robert Sedgewick
- Online: LeetCode Articles (Strings), Tech With Tim YouTube
- Platform: HackerRank (Strings section)

2.2.3 Daily Breakdown

- **Day 1:** Learn string basics and palindrome checks. Solve “Valid Palindrome” (LeetCode Easy).
- **Day 2:** Study KMP algorithm. Solve “Longest Palindromic Substring” (LeetCode Medium).
- **Day 3:** Learn sliding window for strings. Solve “Longest Substring Without Repeating Characters” (LeetCode Medium).
- **Day 4:** Practice string parsing. Solve “Group Anagrams” (LeetCode Medium).
- **Day 5:** Study string compression. Solve “String Compression” (LeetCode Medium).
- **Day 6:** Mixed practice. Solve 2–3 string problems (Easy/Medium).
- **Day 7:** Revise Week 2 problems. Write pseudocode for KMP algorithm.

2.3 Week 3: Linked Lists

2.3.1 Learning Objectives

Understand singly/doubly linked lists, operations, and cycle detection.

2.3.2 Resources

- Book: “Data Structures and Algorithms in Python” by Goodrich
- Online: GeeksforGeeks (Linked Lists), Abdul Bari YouTube
- Platform: LeetCode (Linked List problems)

2.3.3 Daily Breakdown

- **Day 1:** Learn linked list basics and traversal. Solve “Reverse Linked List” (LeetCode Easy).
- **Day 2:** Study merging lists. Solve “Merge Two Sorted Lists” (LeetCode Easy).
- **Day 3:** Learn cycle detection (Floyd’s algorithm). Solve “Linked List Cycle” (LeetCode Easy).
- **Day 4:** Practice list manipulation. Solve “Remove Nth Node From End of List” (LeetCode Medium).
- **Day 5:** Study intersection problems. Solve “Intersection of Two Linked Lists” (LeetCode Easy).
- **Day 6:** Mixed practice. Solve 2–3 linked list problems.
- **Day 7:** Revise Week 3 problems. Explain cycle detection to yourself.

2.4 Week 4: Stacks and Queues

2.4.1 Learning Objectives

Master stack/queue implementations and applications (e.g., monotonic stacks, deque).

2.4.2 Resources

- Book: “Introduction to Algorithms” by Cormen (Chapter 10)
- Online: LeetCode Articles (Stacks), NeetCode YouTube
- Platform: HackerRank (Stacks/Queues)

2.4.3 Daily Breakdown

- **Day 1:** Learn stack basics. Solve “Valid Parentheses” (LeetCode Easy).
- **Day 2:** Study monotonic stacks. Solve “Next Greater Element I” (LeetCode Easy).
- **Day 3:** Learn queue basics and deque. Solve “Sliding Window Maximum” (LeetCode Hard).
- **Day 4:** Practice queue applications. Solve “Design Circular Queue” (LeetCode Medium).
- **Day 5:** Study stack/queue combinations. Solve “Min Stack” (LeetCode Medium).
- **Day 6:** Mixed practice. Solve 2–3 stack/queue problems.
- **Day 7:** Revise Week 4 problems. Re-solve “Min Stack” from scratch.

2.5 Week 5: Trees

2.5.1 Learning Objectives

Understand binary trees, traversals (inorder, preorder, postorder), and BST properties.

2.5.2 Resources

- Book: “Algorithms” by Robert Sedgewick
- Online: GeeksforGeeks (Trees), Tech With Tim YouTube
- Platform: LeetCode (Tree problems)

2.5.3 Daily Breakdown

- **Day 1:** Learn tree traversals. Solve “Binary Tree Inorder Traversal” (LeetCode Easy).

- **Day 2:** Study BST properties. Solve “Validate Binary Search Tree” (LeetCode Medium).
- **Day 3:** Learn tree height calculation. Solve “Maximum Depth of Binary Tree” (LeetCode Easy).
- **Day 4:** Practice LCA problems. Solve “Lowest Common Ancestor of a Binary Tree” (LeetCode Medium).
- **Day 5:** Study path sum problems. Solve “Path Sum” (LeetCode Easy).
- **Day 6:** Mixed practice. Solve 2–3 tree problems.
- **Day 7:** Revise Week 5 problems. Write pseudocode for LCA.

2.6 Week 6: Binary Search

2.6.1 Learning Objectives

Master binary search and its variations (e.g., rotated arrays, finding peaks).

2.6.2 Resources

- Book: “Introduction to Algorithms” by Cormen (Chapter 2)
- Online: LeetCode Articles (Binary Search), NeetCode YouTube
- Platform: HackerRank (Searching)

2.6.3 Daily Breakdown

- **Day 1:** Learn binary search basics. Solve “Binary Search” (LeetCode Easy).
- **Day 2:** Study rotated array search. Solve “Search in Rotated Sorted Array” (LeetCode Medium).
- **Day 3:** Learn peak finding. Solve “Find Peak Element” (LeetCode Medium).
- **Day 4:** Practice range search. Solve “Find First and Last Position of Element in Sorted Array” (LeetCode Medium).
- **Day 5:** Study advanced binary search. Solve “Median of Two Sorted Arrays” (LeetCode Hard).
- **Day 6:** Mixed practice. Solve 2–3 binary search problems.
- **Day 7:** Revise Week 6 problems. Explain binary search variations.

2.7 Week 7: Graphs (BFS/DFS)

2.7.1 Learning Objectives

Understand graph representations, BFS, DFS, and topological sort.

2.7.2 Resources

- Book: “Algorithms” by Robert Sedgewick
- Online: GeeksforGeeks (Graphs), Abdul Bari YouTube
- Platform: LeetCode (Graph problems)

2.7.3 Daily Breakdown

- **Day 1:** Learn graph representations (adjacency list/matrix). Solve “Number of Islands” (LeetCode Medium).
- **Day 2:** Study BFS. Solve “Rotting Oranges” (LeetCode Medium).
- **Day 3:** Learn DFS. Solve “Course Schedule” (LeetCode Medium).
- **Day 4:** Practice topological sort. Solve “Course Schedule II” (LeetCode Medium).
- **Day 5:** Study flood fill. Solve “Flood Fill” (LeetCode Easy).
- **Day 6:** Mixed practice. Solve 2–3 graph problems.
- **Day 7:** Revise Week 7 problems. Re-solve “Course Schedule” from scratch.

2.8 Week 8: Advanced Graph Algorithms

2.8.1 Learning Objectives

Master Dijkstra’s, Kruskal’s, and Union-Find algorithms.

2.8.2 Resources

- Book: “Introduction to Algorithms” by Cormen (Chapter 24)
- Online: GeeksforGeeks (Graph Algorithms), NeetCode YouTube
- Platform: LeetCode (Graph problems)

2.8.3 Daily Breakdown

- **Day 1:** Learn Dijkstra’s algorithm. Solve “Network Delay Time” (LeetCode Medium).
- **Day 2:** Study Kruskal’s algorithm. Solve “Min Cost to Connect All Points” (LeetCode Medium).
- **Day 3:** Learn Union-Find. Solve “Number of Connected Components in an Undirected Graph” (LeetCode Medium).
- **Day 4:** Practice shortest path problems. Solve “Cheapest Flights Within K Stops” (LeetCode Medium).

- **Day 5:** Study bipartite graphs. Solve “Is Graph Bipartite?” (LeetCode Medium).
- **Day 6:** Mixed practice. Solve 2–3 graph problems.
- **Day 7:** Revise Week 8 problems. Write pseudocode for Dijkstra’s.

2.9 Week 9: Dynamic Programming (DP)

2.9.1 Learning Objectives

Understand 1D/2D DP, memoization, and tabulation.

2.9.2 Resources

- Book: “Dynamic Programming for Coding Interviews” by Meenakshi
- Online: LeetCode Articles (DP), Tech With Tim YouTube
- Platform: LeetCode (Dynamic Programming)

2.9.3 Daily Breakdown

- **Day 1:** Learn 1D DP. Solve “Climbing Stairs” (LeetCode Easy).
- **Day 2:** Study memoization. Solve “House Robber” (LeetCode Medium).
- **Day 3:** Learn 2D DP. Solve “Longest Common Subsequence” (LeetCode Medium).
- **Day 4:** Practice knapsack problems. Solve “0/1 Knapsack” (GeeksforGeeks).
- **Day 5:** Study advanced DP. Solve “Edit Distance” (LeetCode Hard).
- **Day 6:** Mixed practice. Solve 2–3 DP problems.
- **Day 7:** Revise Week 9 problems. Explain DP state transitions.

2.10 Week 10: Advanced DP

2.10.1 Learning Objectives

Master complex DP problems (e.g., matrix paths, stock trading).

2.10.2 Resources

- Book: “Introduction to Algorithms” by Cormen (Chapter 15)
- Online: GeeksforGeeks (DP), NeetCode YouTube
- Platform: LeetCode (Dynamic Programming)

2.10.3 Daily Breakdown

- **Day 1:** Learn matrix DP. Solve “Unique Paths” (LeetCode Medium).

- **Day 2:** Study stock trading problems. Solve “Best Time to Buy and Sell Stock” (LeetCode Easy).
- **Day 3:** Practice advanced stock problems. Solve “Best Time to Buy and Sell Stock with Cooldown” (LeetCode Medium).
- **Day 4:** Learn interval DP. Solve “Burst Balloons” (LeetCode Hard).
- **Day 5:** Study bitmask DP. Solve “Partition Equal Subset Sum” (LeetCode Medium).
- **Day 6:** Mixed practice. Solve 2–3 DP problems.
- **Day 7:** Revise Week 10 problems. Re-solve “Burst Balloons” from scratch.

2.11 Week 11: Advanced Topics

2.11.1 Learning Objectives

Understand tries, segment trees, and bit manipulation.

2.11.2 Resources

- Book: “Algorithms” by Robert Sedgewick
- Online: GeeksforGeeks (Advanced DS), Abdul Bari YouTube
- Platform: LeetCode (Trie, Bit Manipulation)

2.11.3 Daily Breakdown

- **Day 1:** Learn trie basics. Solve “Implement Trie (Prefix Tree)” (LeetCode Medium).
- **Day 2:** Study trie applications. Solve “Word Search II” (LeetCode Hard).
- **Day 3:** Learn bit manipulation. Solve “Single Number” (LeetCode Easy).
- **Day 4:** Practice advanced bit manipulation. Solve “Number of 1 Bits” (LeetCode Easy).
- **Day 5:** Study segment trees. Solve “Range Sum Query - Mutable” (LeetCode Medium).
- **Day 6:** Mixed practice. Solve 2–3 advanced problems.
- **Day 7:** Revise Week 11 problems. Write pseudocode for trie operations.

2.12 Week 12: Mock Interviews and Mixed Practice

2.12.1 Learning Objectives

Simulate real interviews and solve mixed problem sets.

2.12.2 Resources

- Platforms: Pramp, Interviewing.io, LeetCode Premium
- Online: NeetCode (Mock Interview Videos)

2.12.3 Daily Breakdown

- **Day 1:** Solve mixed problems (1 Easy, 1 Medium, 1 Hard) from any topic.
- **Day 2:** Conduct a mock interview on Pramp or Interviewing.io.
- **Day 3:** Solve company-specific problems (e.g., Google, Amazon tagged on LeetCode).
- **Day 4:** Practice system design basics (if applicable). Review DSA problems.
- **Day 5:** Solve 2–3 Hard problems from mixed topics.
- **Day 6:** Conduct another mock interview. Analyze feedback.
- **Day 7:** Revise weak areas. Solve 3–4 problems from identified topics.

3 Daily 3-Hour Structure

3.1 Hour 1: Learning

- Study concepts using recommended resources.
- Take notes on algorithms, complexities, and edge cases.
- Example: For graphs, learn BFS implementation and analyze its $O(V+E)$ complexity.

3.2 Hour 2–2.5: Practice

- Solve 2–3 problems daily:
 - Weeks 1–4: 2 Easy, 1 Medium
 - Weeks 5–8: 1 Easy, 2 Medium
 - Weeks 9–12: 1 Medium, 1–2 Hard
- Approach:
 1. Read and understand the problem (5–10 mins).
 2. Plan solution or write pseudocode (5 mins).
 3. Code and test with edge cases (15–20 mins).
 4. Optimize and analyze complexity (5–10 mins).

- If stuck, spend 20 minutes before checking discussions or editorials.

3.3 Hour 2.5–3: Revision

- Revisit problems solved 2–3 days ago.
- Re-solve without looking at previous code to test retention.
- Use a spreadsheet or Anki to track problems and concepts.
- Example: Re-solve a DP problem and explain the state transition.

4 Tips for Success

- **Consistency:** Dedicate 3 hours daily, including weekends.
- **Tracking:** Log problems in a spreadsheet with columns for topic, difficulty, and mistakes.
- **Mock Interviews:** Start weekly mock interviews from Week 8 on platforms like Pramp or Interviewing.io.
- **Community:** Join DSA groups on Discord, Reddit, or X (<https://x.com>) for support.
- **Soft Skills:** Prepare behavioral answers and basic system design (for senior roles).
- **Health:** Take short breaks every hour to avoid burnout.

5 Conclusion

This 12-week, 3-hour daily DSA routine provides a structured path to master the skills needed for a 50 LPA offer. By focusing on learning, practicing, and revising consistently, you will build confidence and proficiency in solving complex problems. Stay disciplined, seek feedback, and good luck in your interview preparation!